

LIMITS & CONTINUITY

Limits (THEORY)

Type I

$$\lim_{x \rightarrow a} f(x) = f(a) \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{f(a)}{g(a)}; \text{ if } g(a) \neq 0$$

Type II

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ & $g(a) = 0$, then cancel the common terms from numerator and denominator using algebraic treatments.

The reduced form would be: $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)}$

Type III

$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$, Divide numerator and denominator by the highest power of x , and then put $1/x = 0$.

Type IV(Standard Limits)

- $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \quad \lim_{x \rightarrow 0} \frac{e^{mx} - 1}{x} = m \quad \lim_{x \rightarrow 0} \frac{e^{mx} - 1}{mx} = 1$
- $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a \quad \lim_{x \rightarrow 0} \frac{a^{mx} - 1}{x} = m \cdot \log_e a \quad \lim_{x \rightarrow 0} \frac{a^{mx} - 1}{mx} = \log_e a$
- $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \quad \lim_{x \rightarrow 0} \frac{\log(1+mx)}{x} = m \quad \lim_{x \rightarrow 0} \frac{\log(1+mx)}{mx} = 1$
- $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n \cdot a^{n-1} \quad \lim_{x \rightarrow a} \frac{x^n - a^n}{x^m - a^m} = \frac{n \cdot a^{n-1}}{m \cdot a^{m-1}} = \frac{n}{m} \cdot a^{n-m}$
- $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e \quad \lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x = e^a$
- $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e; \lim_{x \rightarrow \infty} (1+x)^{\frac{a}{x}} = e^a; \lim_{x \rightarrow 0} (1+ax)^{\frac{1}{x}} = e^a$

CONCEPT OF CONTINUITY OF A FUNCTION

A function $f(x)$ is said to be Continuous at a particular point, $x = a$, if it satisfy the following conditions:

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$$

Left hand = Right hand = Functional
Limit (LHL) Limit (RHL) Value

Note1: Equality of RHL and LHL is treated as a condition for existence of limit i.e, limit of a function will exist if LHL=RHL

Note2: For Continuity, equality of the functional value at that point is also necessary.

Note3: For all Continuous functions, limit must exist, but existence of limit, is not a sufficient condition for continuity of a function.

Note4: Sum, difference, product and quotient of all continuous functions are always continuous.

Note5: All polynomials are continuous.

Note6: If a given function is of the form $\frac{f(x)}{g(x)}$, where both $f(x)$ and $g(x)$ are polynomials in x , it will be everywhere continuous except at the points at which it is undefined i.e; points of discontinuity of such functions are the points where $g(x) = 0$.

Example: In each of the following cases, discuss continuity of the functions at $x=5$

i) $f(x) = \frac{x^2 - 25}{x - 5}$

Solution: LHL = $\lim_{x \rightarrow 5^-} \frac{x^2 - 25}{x - 5} = \lim_{x \rightarrow 5^-} \frac{2x}{1} = 2 \times 5 = 10$

$$\text{RHL} = \lim_{x \rightarrow 5^+} \frac{x^2 - 25}{x - 5} = \lim_{x \rightarrow 5^+} \frac{2x}{1} = 2 \times 5 = 10$$

$$f(5) = \frac{25-25}{5-5} = \frac{0}{0} (\text{undefined})$$

since, $\text{LHL} = \text{RHL} \neq f(5)$, $f(x)$ is discontinuous at $x = 5$, although the limit has existed.

$$\text{ii) } f(x) = \frac{x^2 - 25}{x - 5}, \text{ when } x \neq 5$$

$$= 10, \text{ when } x = 5$$

Solution: $\text{LHL} = 10 = \text{RHL}$ taken from (i)

Given, $f(5) = 10$ since, $\text{LHL} = \text{RHL} = f(5)$, $f(x)$ is continuous at $x = 5$

$$\text{iii) } f(x) = \frac{x^2 - 25}{x - 5}, \text{ when } x \neq 5$$

$$= 2, \text{ when } x = 5$$

Solution: $\text{LHL} = \text{RHL} = 10$ taken from (ii)

Given, $f(5) = 2$ since, $\text{LHL} = \text{RHL} \neq f(5)$, $f(x)$ is discontinuous at $x = 5$

Example 2: Find the points of discontinuity of the function, $f(x) = \frac{(x^2 - 3x + 2)}{(x^2 - 5x + 6)}$

Solution: The given function will be continuous at all points, except at the points at which it is undefined i.e the points at which its denominator is 0. $(x^2 - 5x + 6) = 0$

Points of discontinuity are 2 and 3

$$\Rightarrow (x - 2)(x - 3) = 0$$

$$\Rightarrow x = 2, 3$$

WORKING CODES for Q. No. 1 to 18

Mark C : if function is continuous at the given point

Mark D : if function is discontinuous at the given point

Mark X : if nothing can be said about the continuity of the function at the given point

Mark Y : if function is neither continuous nor discontinuous at the given point